Task 10 — Comparative Insights

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST PlaceMux · Phase 1 · Study Guide

DIFFICULTY **Building · Proficient** Level 10 of 20 · ramp 48%

1 · What this is & why it matters

Produce comparative insights: segment, cohort and period comparisons that explain why numbers differ.

A total hides the story. Comparisons reveal which segment, region or cohort is driving the result.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A comparative analysis identifying the segments driving the result.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Cohorting

grouping by a shared start point to compare like with like.

Segmentation

slicing by attributes (region, plan, channel).

Like-for-like

controlling for size/time so comparisons are fair.

Contribution analysis

which segment moved the total, and by how much.

Simpson's paradox

aggregate trends can reverse within segments; always check.

5 · Recommended stack & tools

- pandas groupby/pivot
- seaborn
- a comparison table

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Define meaningful segments/cohorts for the question.
2. Compute the metric within each and compare.

3. Normalise for size so big segments don't dominate unfairly.
4. Attribute the overall change to the driving segments.
5. Check for Simpson's paradox before concluding.
6. Summarise who is up, who is down, and why.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Absolute vs indexed comparisons.
- Cohort tables vs small-multiple charts.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- Is the headline trend hiding an opposite trend in a key segment?
- Which comparison would change the recommendation?
- Are the segments fair, or confounded by something else?

9 · Definition of Done

Done when all are true and demoable

- A comparative analysis identifying the segments driving the result.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A comparative analysis identifying the segments driving the result.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Comparing unequal-sized groups raw.
- Missing a reversing sub-trend.
- Cherry-picking the flattering comparison.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design
- Time-series and segmentation techniques